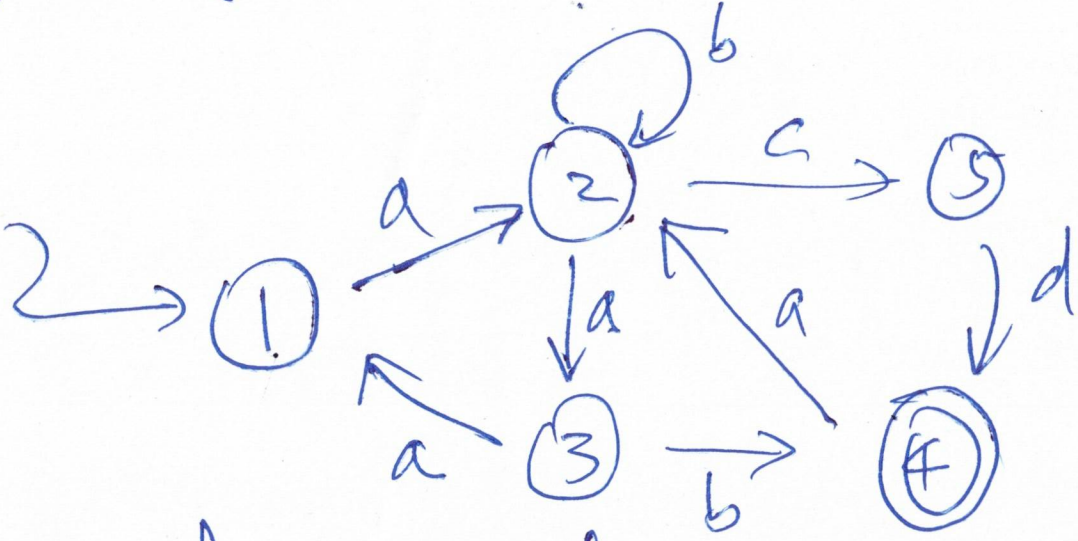
5/1 CptS 350

Final Exam Review III.

local propagation (not in the Exam)

Example:

DFA



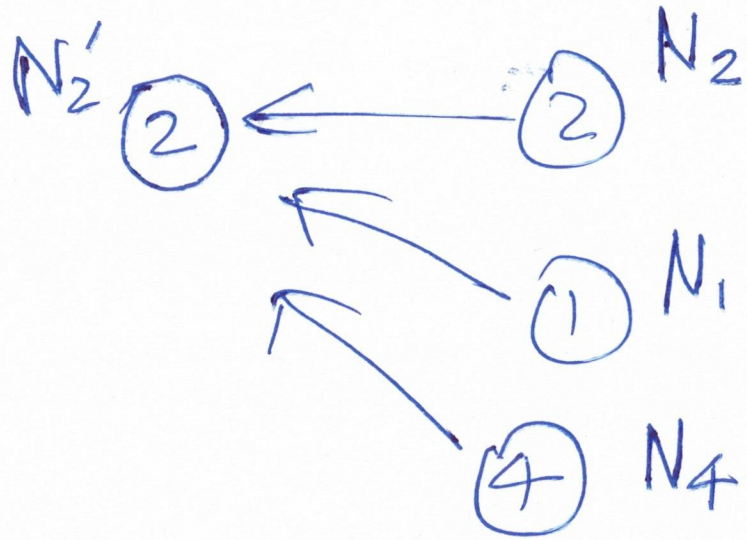
Want: # of walks from ① to ④ with len = 10. (total # of words accepted with len = 10).

Each node i is associated with

N_i , N_i'

(old #) (new #)

Key: you need tell me how to update the counts.



the new # N_2' is updated from old counts

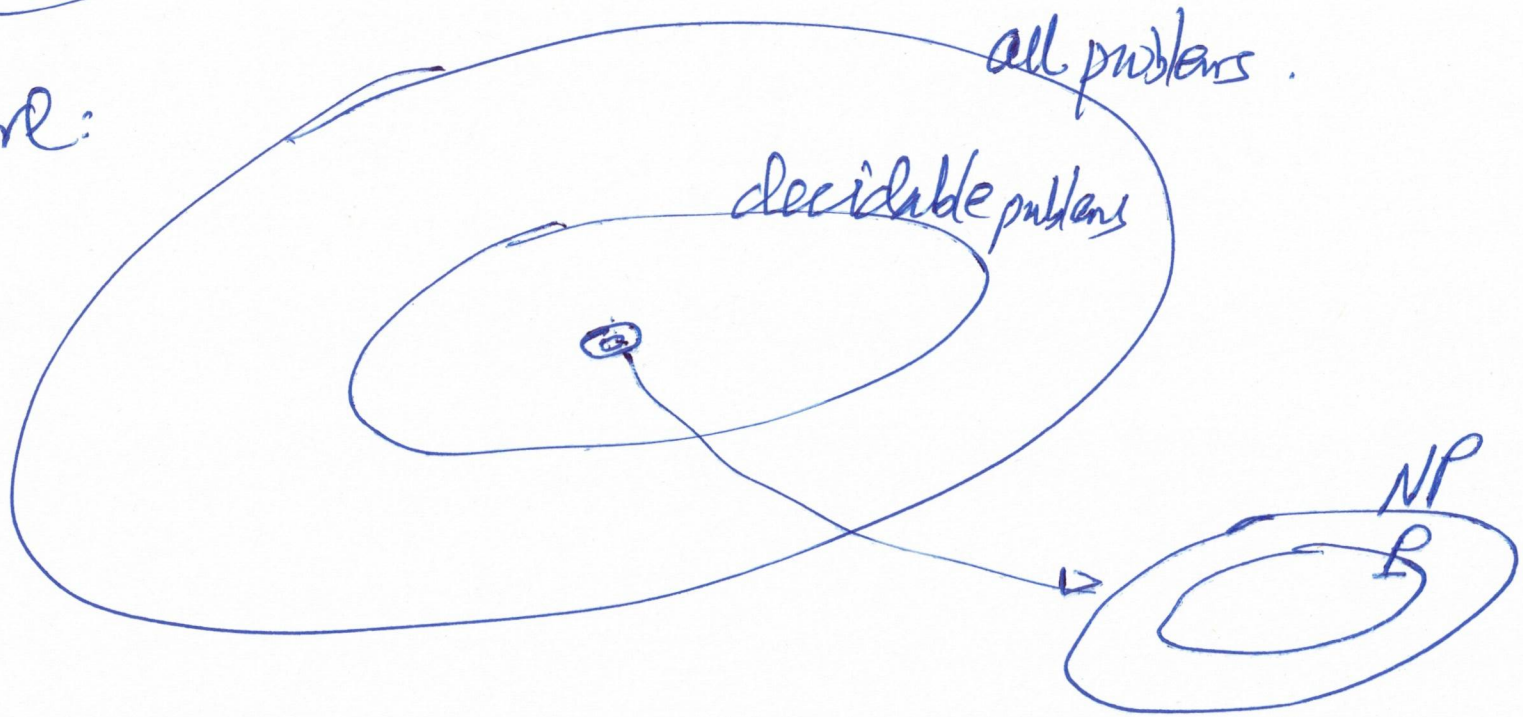
N_2, N_1, N_4.

Key: Alg is to update <u>all</u> the counts on all nodes <u>in parallel</u> !

Today: P, NP, NP-complete. hw hint.

Type your cheat sheets using a small font. you do need put a lot of things down.

Big picture:



P ≠ NP.

you must know the basic definitions:

P: The class of problem that can be solved by (det) algs in poly-time;

NP: " " " " " "

by nondet algs in poly-time;

$P \subseteq NP$; But $P \stackrel{?}{=} NP$.

$A \leq_m B$ is defined by:

there is a poly-time computable function f s.t. $\forall x$, $x \in A$ iff $f(x) \in B$.

A is NP-complete if

① $A \in NP$

② $\forall B \in NP,\ B \leq_m A$.

A is NP-hard if $\forall B \in NP,\ B \leq_m A$.

How to write proofs ?

How to prove a problem is in NP (using Guess-and-check template).

Example: I define a problem $\mathcal{CD}$ as below:

Given: a graph G, a number k

Question: Is there a set of k nodes in G s.t. for any two nodes V, V', in the set, $\langle V, V' \rangle$ is not an edge in G?

Show: $\mathcal{CD} \in NP$.

Proof. I design a nondet. poly-time alg to solve co using guess-check template:

input: G, k

guess: a set of k nodes in G // size of guess $\leq$ size of graph.

check: for any two nodes v, v' in the set, $\langle v, v' \rangle$ is not an edge in G.

runs in det. poly-time

if checking is okay
    say yes.

crash.

Clearly, the nondet alg runs in poly-time. ▨.

More proofs ( Work on it!)

(See class notes)

1. $A \leq_m B$, $B \leq_m C \Rightarrow A \leq_m C$. ✓ in class

2. $A \leq_m B$, $B \in P \Rightarrow A \in P$.

3. $A \leq_m B$, $B \in NP \Rightarrow A \in NP$

4. $A \leq_m B \Rightarrow \overline{A} \leq_m \overline{B}$

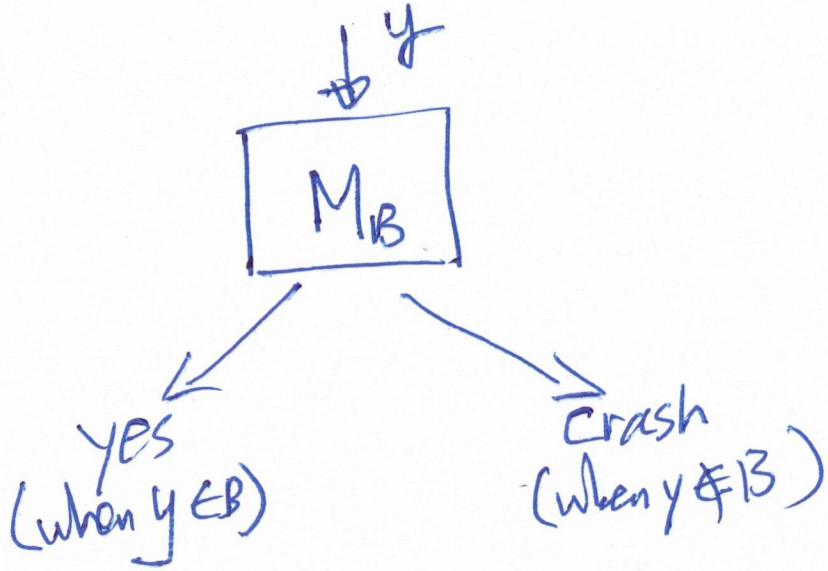
↳ complement of A.

Show: $A \leq_m B$, $B \in NP \Rightarrow A \in NP$.

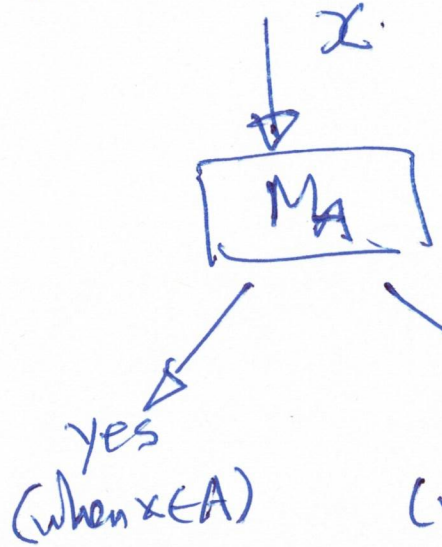
Proof. Since $A \leq_m B$, we have a poly-time computable function f s.t. $\forall x$

$x \in A$ iff $f(x) \in B$.

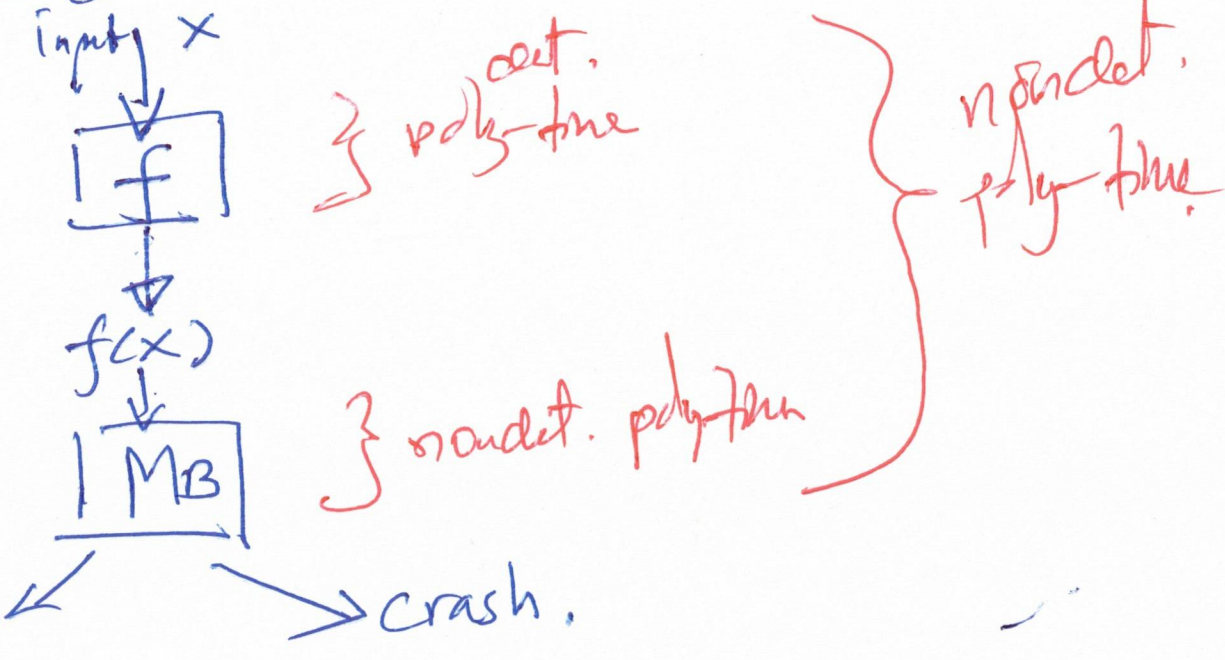
Since $B \in NP$, we have a nondet. poly-time alg M_B to solve B:



To show $A \in NP$, we need construct a nondet poly-time alg M_A :



Clearly, M_A can be constructed as :



hw Hint:

p2. (use guess-check template). I design a nondet. poly-time alg to solve the problem using guess-check template:

input G

guess a path with len ≤ # of nodes in G.

// size of guess ≤ // size of input.

det. poly-time, { check: every node in G is covered in the path exactly once and the path is indeed a path in G

if checking is okay, set yes

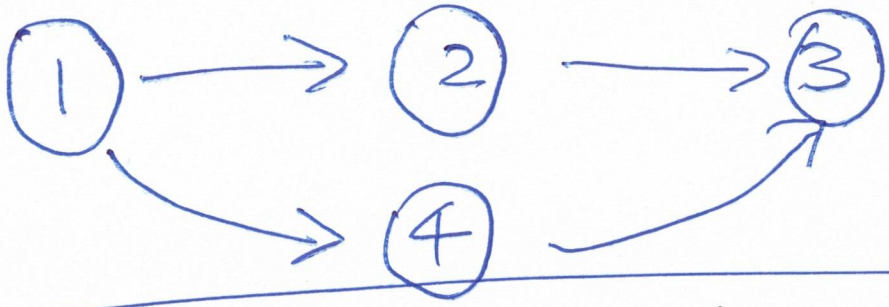
crash.

P3. It's a hard problem! I will show the problem $\in P$.

Example:



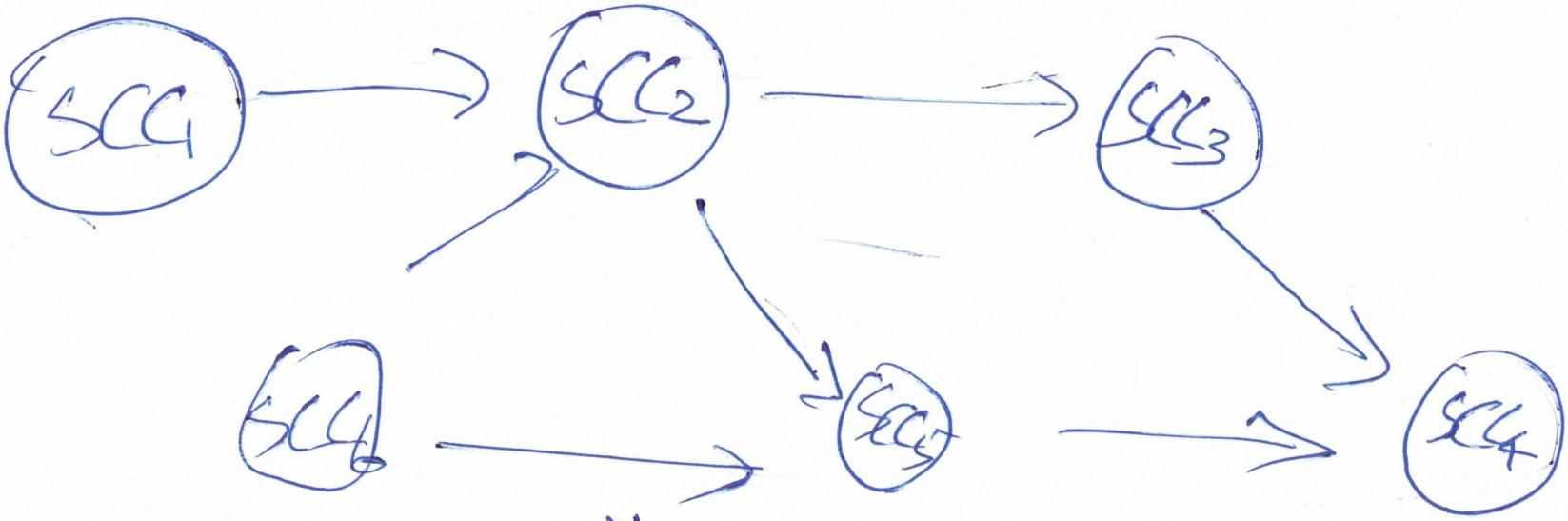
✓



no

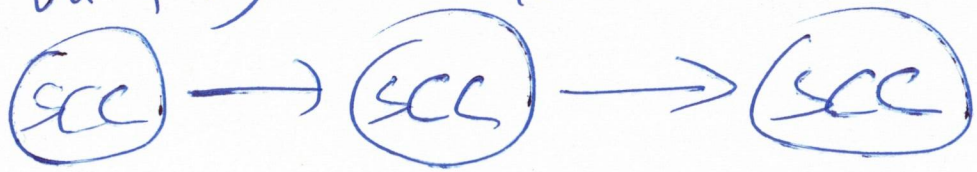
Hint: I run SCC on G. I will obtain a DAG where each node (BigNode) is a SCC of G:

Example:



this DAG will NOT give a yes answer to the problem.

Summary: In order to have a walk in G that covers every node in G, we need make sure that the DAG (obtained from scc) in BigNodes is a "line".



P4. C is a Bool fnctn with

$$y = C(x_1 \ldots x_k).$$

C is satisfiable if there are bits $x_1 \ldots x_k$ s.t. $C(x_1 \ldots x_k) = y = 1$.

Assume that we have a det. poly-time alg M_{SAT} that decides whether C is sat.

Define $C_1 \equiv C_2$ if $\forall x_1 \ldots x_k$,

$$C_1(x_1 \ldots x_k) = C_2(x_1 \ldots x_k).$$

To show: we have det. poly-time alg to decide $C_1 \equiv C_2$.

Proof. $C_1 \not\equiv C_2$ iff $\exists x_1 \cdots x_n$

$C_1(x_1 \cdots x_n) \neq C_2(x_1 \cdots x_n)$

iff $\exists x_1 \cdots x_n$.

$(C_1(x_1 \cdots x_n) \wedge \overline{C_2}(x_1 \cdots x_n)) \vee$

$(\overline{C_1}(x_1 \cdots x_n) \wedge C_2(x_1 \cdots x_n))$

iff $((C_1 \wedge \overline{C_2}) \vee (\overline{C_1} \wedge C_2))$ is satisfiable.

can be solved by MSAT in poly-time.

□.

Cpt S 350 Homework #10

Please print your name!

1. Describe a proof that, for any three NP-problems A, B, C, we have $A \leq_m B$ and $B \leq_m C$ implies $A \leq_m C$.

2. Show that the following problem is in NP (that is, you need only describe a nondeterministic polynomial-time algorithm that solves the following problem):

   Given: a directed graph G,

   Question: is there a path on G such that every node of G is covered exactly once?

3. Show that the following problem is in NP:

   Given: a directed graph G,

   Question: is there a path on G such that every node of G is covered?

4. Let C be a Boolean circuit (using AND, NOT, OR gates), which has input $(x_1, \cdots, x_n)$ and one output y. The circuit is satisfiable if for some input, the output y produced by C is 1. Suppose that we have a deterministic polynomial time algorithm that decides whether C is satisfiable.

   Now, let C_1 and C_2 be two Boolean circuits (using AND, NOT, OR gates), each of which has input $(x_1, \cdots, x_n)$ and one output y. We say that the two circuits are equivalent if for any input, the output produced by C_1 equals the the output produced by C_2.

   Show that we also have a deterministic polynomial time algorithm that decides whether C_1 and C_2 are equivalent.

Well known NP-complete problems:

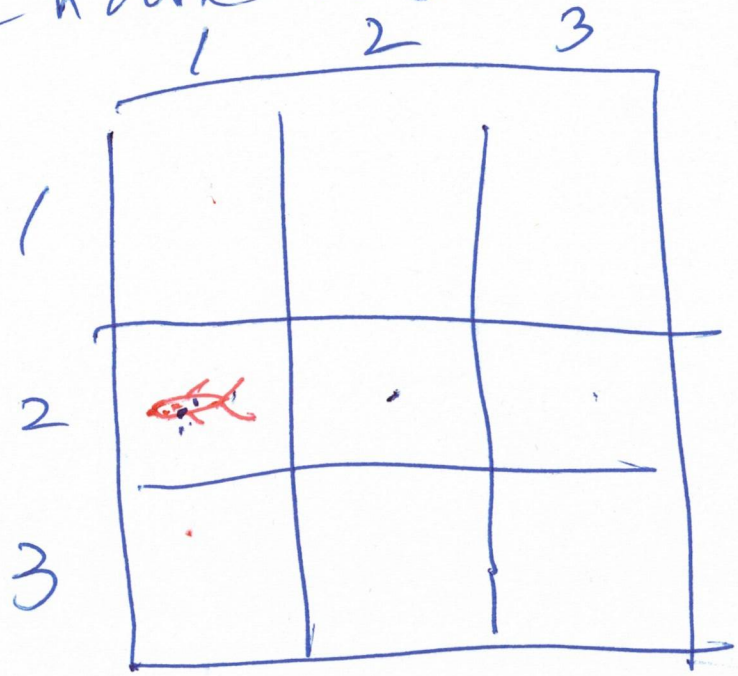
SAT, 3SAT, IDSET, VC, HC.

We do have problems that we don't if they are NP-complete: GI, FACT.

Hard problems: graph = (chess) board

I have a chess board of k x k size.

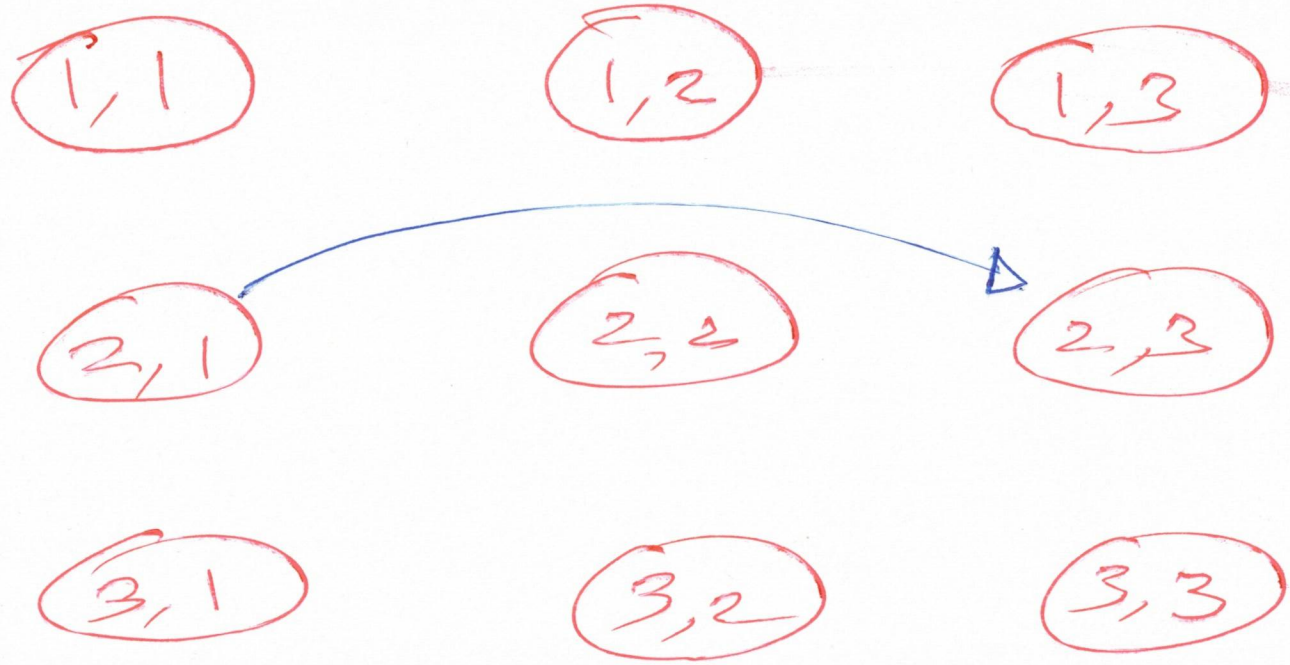
(3 x 3).



I have a piece which can move in straight line (vertically, horizontally).

Encode the board to a graph.

What are the nodes in the graph?



A "move" is an edge in the graph. Example:

 Can move from position (2,1) to (2,3)

In this way, you can solve a problem on the board through a graph search.